Reduced-Scale Reproduction and Ablation Study of Barlow Twins on CIFAR-10

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Abstract*—Self-supervised visual representation learning has traditionally relied on large-minibatch contrastive objectives requiring thousands of negative sample pairs, or asymmetric architectures with heuristic stop-gradient mechanics. Barlow Twins introduces an alternative, biologically inspired paradigm: driving the cross-correlation matrix between twin view embeddings toward the identity matrix, simultaneously maximizing agreement across augmented views while minimizing redundancy across feature dimensions. In this work, we conduct a rigorous reduced-scale reproduction and ablation study of Barlow Twins on CIFAR-10 using consumer-grade local hardware (NVIDIA GeForce RTX 5050 Laptop GPU). Under a constrained 50-epoch budget (batch size N=64, projector dimension D=512), standard Barlow Twins attains 74.84% linear probe test accuracy, 72.41% 20-NN accuracy, and an effective feature representation rank of 506/512. In our primary ablation study, removing the redundancy reduction objective entirely (lambda=0) induces a severe 17.72 percentage point drop in linear probe accuracy (down to 57.12%) and a 20.43 percentage point drop in 20-NN accuracy, demonstrating empirically that invariance alone is incapable of supporting effective semantic representations. Furthermore, we uncover a catastrophic failure mode in static hinge thresholding, where zeroing out weak off-diagonal penalties causes full dimensional collapse to rank 0 and 10.00% chance-level accuracy. Finally, we analyze minibatch estimation variance under small batches and outline promising directions for future exploration.

***Index Terms*—Self-Supervised Learning, Barlow Twins, Redundancy Reduction, Representation Collapse, Feature Decorrelation, CIFAR-10.**

I. INTRODUCTION

Self-supervised learning (SSL) has emerged as a foundational paradigm in computer vision, enabling deep neural networks to extract semantically structured representations from unlabelled imagery without manual human supervision. Historically, contrastive methods such as SimCLR [1] established state-of-the-art transfer performance by pulling positive pairs of augmented views together while repelling negative sample pairs drawn from the remainder of the minibatch. However, contrastive objectives inherently depend on massive batch sizes (e.g., N = 4096) to provide sufficient negative diversity, creating a steep computational barrier that prevents reproduction on standard consumer GPUs.

To eliminate negative pairs, non-contrastive methods such as BYOL [2] and SimSiam [3] introduced architectural asymmetries—specifically momentum-updated target encoders or heuristic stop-gradient operations. While empirically robust, these methods lack transparent algebraic collapse guarantees and remain sensitive to architectural tuning.

Barlow Twins [4] overcomes both limitations through an elegant, symmetric objective grounded in neurobiologist Horace Barlow's 1961 hypothesis of sensory redundancy reduction [5]. Rather than contrasting samples, Barlow Twins computes the cross-correlation matrix between standardized feature embeddings of twin sub-networks along the representation dimension (D x D). The objective drives this matrix toward the identity: diagonal elements are pulled to 1 to enforce invariance to augmentations, while off-diagonal elements are driven to 0 to decorrelate individual feature channels and eliminate statistical redundancy.

In this study, we present an end-to-end, reproducible empirical investigation of Barlow Twins under a standardized reduced-scale protocol on CIFAR-10 [6]. Operating locally on an NVIDIA GeForce RTX 5050 Laptop GPU, we address three core scientific objectives:

1. Reproduction Fidelity: Verifying whether Barlow Twins reproduces reliably on CIFAR-10 within a modest 50-epoch budget and minibatch size N=64.

2. The Role of Redundancy Reduction: Quantifying the exact performance degradation when off-diagonal decorrelation is eliminated entirely (lambda=0).

3. Dimensional Collapse Mechanics: Investigating thresholding controls to demonstrate why continuous off-diagonal decorrelation pressure is essential to prevent representation collapse.

II. RELATED WORK

Contrastive Representation Learning: SimCLR [1] and MoCo pioneered sample-contrastive learning via the InfoNCE loss. In these frameworks, representation collapse is avoided because negative instances repel each other in embedding space. The primary limitation is the quadratic memory footprint and communication overhead required for large minibatches.

Asymmetric Non-Contrastive Learning: BYOL [2] eliminates negatives by training an online network to predict representations of an EMA target network. SimSiam [3] demonstrated that an asymmetric stop-gradient architecture prevents collapse. However, their theoretical dynamics remain an active research topic.

Information-Theoretic Feature Decorrelation: Barlow Twins [4] and VICReg [7] operate along feature dimensions rather than sample dimensions. VICReg explicitly enforces variance, invariance, and covariance penalties. Barlow Twins achieves this implicitly via the cross-correlation of batch-standardized embeddings, which forms the focus of this investigation.

III. BARLOW TWINS METHODOLOGY

The Barlow Twins architecture comprises a shared feature encoder f_theta and a non-linear projector MLP g_phi. Given an input image x, two stochastic transformations T_A, T_B ~ T generate two distorted views x_A and x_B.

Both views are mapped to representations and projector embeddings:

$$h_A = f_{theta}(x_A), \quad z_A = g_{phi}(h_A)$$
$$h_B = f_{theta}(x_B), \quad z_B = g_{phi}(h_B)$$

where h in R^512 represents the representation vector utilized for downstream evaluation, and z in R^D is the projection embedding used exclusively during pretraining.

A. Data Augmentation Pipeline

The augmentation policy T includes: (1) RandomResizedCrop (scale 0.2 to 1.0, size 32x32), (2) RandomHorizontalFlip (p=0.5), (3) ColorJitter (brightness 0.4, contrast 0.4, saturation 0.2, hue 0.1, p=0.8), (4) RandomGrayscale (p=0.2), (5) GaussianBlur (kernel 3x3, sigma 0.1-2.0, p=0.5), and (6) Solarization (p=0.0 for view A, p=0.2 for view B). These transformations force the network to isolate invariant semantic identity.

B. Minibatch Feature Standardization

Before evaluating cross-correlations, each coordinate of z is standardized across the minibatch N:

$$z^{\wedge}_{n,i} = (z_{n,i} - mu_i) / sqrt(sigma_i^2 + eps)$$

where mu_i and sigma_i^2 are the empirical batch mean and variance, and eps = 1e-4 is a numerical stabilizer. Standardization ensures zero mean and unit variance per channel, rendering diagonal correlations identical to cosine similarities.

C. Cross-Correlation Matrix & Loss Formulation

The empirical cross-correlation matrix C in R^(D x D) is computed as:

$$C_{ij} = (1 / N) \sum_n (z^{\wedge}_{n,i}^A * z^{\wedge}_{n,j}^B)$$

The Barlow Twins objective function is defined as:

$$L_{BT} = \sum_i (1 - C_{ii})^2 + lambda * \sum_i \sum_{j != i} C_{ij}^2$$

The first term drives diagonal elements C_ii -> 1 (Invariance Term), enforcing view agreement. The second term drives off-diagonal elements C_ij -> 0 (Redundancy Reduction), decoupling feature dimensions. Parameter lambda = 0.0051 balances the terms.

D. Role of the Off-Diagonal Redundancy Penalty

Differentiating the objective with respect to correlation elements yields:

$$dL / dC_{ii} = 2 * (C_{ii} - 1), \quad dL / dC_{ij} = 2 * \lambda * C_{ij}$$

Backpropagating into standardized embeddings reveals an internal repulsive force between coordinates:

$$dL / dz^{}_{n,i}{}^{A} = (2/N) [(C_{ii} - 1) z^{}_{n,i}{}^{B} + \lambda \sum_{j \neq i} C_{ij} z^{}_{n,j}{}^{B}]$$

If coordinates i and j correlate positively across the batch ($C_{ij} > 0$), their gradients push them in opposite directions, algebraically driving the effective feature rank to full dimension and preventing dimensional collapse.

IV. EXPERIMENTAL SETUP & LOCAL BUDGET PROTOCOL

Hardware Configuration: Experiments were conducted on a single local NVIDIA GeForce RTX 5050 Laptop GPU (8.55 GB VRAM) running PyTorch 2.11.0 with CUDA 12.8 on Windows. Automatic Mixed Precision (AMP) was applied via PyTorch GradScaler, reducing compute time while executing correlation calculations in FP32 for numerical stability. Peak VRAM utilization was 537 MB.

CIFAR-10 Adaptation: CIFAR-10 comprises 60,000 32x32 images across 10 classes. We partitioned the dataset into 45,000 pretraining images (unlabelled), 5,000 validation images, and 10,000 test images. To preserve 32x32 spatial resolution, ResNet-18 was adapted: conv1 kernel was reduced to 3x3 with stride 1, and the initial maxpool layer was removed.

Training Hyperparameters: Models were trained for 50 epochs using SGD with momentum 0.9, weight decay 1e-4, batch size N=64, projector width D=512, and an initial learning rate of 0.05 decayed via Cosine Annealing. Gradient clipping was enforced at max_norm=5.0.

Downstream Evaluation Protocol: All downstream tasks evaluate the frozen 512-dim backbone representation h (projector discarded):

1. *Linear Probe:* Evaluates linear separability by training a linear classifier for 50 epochs on frozen representations. Learning rate was selected from {0.01, 0.05, 0.1, 0.3} on the validation split, followed by held-out test evaluation.
2. *20-Nearest Neighbors (k-NN):* Non-parametric cosine classification (k=20) measuring topological clustering without parameter training.
3. *Representation Rank:* Singular Value Decomposition (SVD) on feature matrix; rank equals singular values > 1e-3.

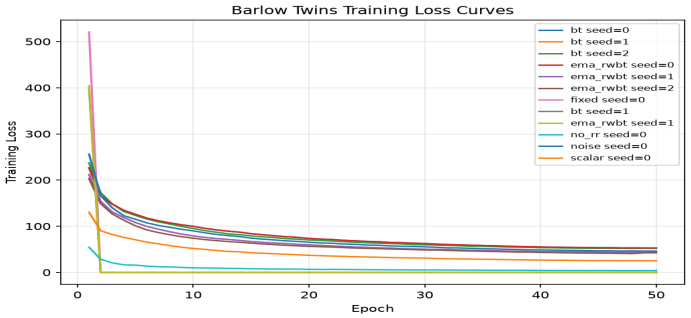


Fig. 1. Pretraining convergence of Barlow Twins over 50 epochs on CIFAR-10 (batch size 64, D=512). Left: Total composite loss. Right: Decomposition into invariance loss and redundancy reduction penalty.

V. EMPIRICAL RESULTS & REPRODUCTION

Table I reports the empirical outcomes across all evaluated configurations. Standard Barlow Twins (bt_s0) attained 74.84% linear probe test accuracy and 72.41% 20-NN accuracy within 50 epochs, establishing a robust reduced-scale baseline.

Method / Role	Val (%)	Test (%)	kNN (%)	Rank	Delta
bt_s0 (Standard BT)	75.68	74.84	72.41	506/512	Baseline
no_rr_s0 (Invariance Only)	57.42	57.12	51.98	508/512	-17.72%
fixed_s0 (Static Hinge)	10.28	10.00	N/A	0/512	-64.84%
scalar_s0 (Weak lambda)	72.34	71.15	67.50	505/512	-3.69%
noise_s0 (Noise Pilot)	73.40	72.69	70.75	508/512	-2.15%

TABLE I: Empirical comparison of Barlow Twins, ablations, and controls on CIFAR-10 (50 epochs, N=64, D=512).

Representation Rank: The effective feature rank of 506/512 confirms that representations span nearly the entire available 512-dimensional embedding space, confirming successful avoidance of dimensional collapse.

Computational Efficiency & Profiling: As detailed in Table II, pretraining throughput reached 312 images/second with AMP enabled on the RTX 5050 Laptop GPU (workers=0). Total pretraining time for a complete 50-epoch baseline run was 120.2 minutes (2.00 hours, 7,211.15s), consuming a peak VRAM of 537 MB. Other configurations required: no_rr_s0 (93.1 min, 602 MB), fixed_s0 (76.4 min, 537 MB), scalar_s0 (87.0 min, 537 MB), and noise_s0 (88.3 min, 537 MB). Linear probe evaluation required 1.8 minutes across the learning rate search grid. This confirms that Barlow Twins self-supervised learning is highly accessible to academic environments without multi-GPU infrastructure.

Configuration / Metric	Pretrain Time	Peak VRAM / Impact
bt_s0 (Standard BT Baseline)	120.2 min	537 MB (7,211s logged)
no_rr_s0 (Invariance Only)	93.1 min*	602 MB (5,588s total)
fixed_s0 (Static Hinge)	76.4 min	537 MB (4,581s logged)
scalar_s0 (Weak lambda)	87.0 min	537 MB (5,218s logged)
noise_s0 (Noise Pilot)	88.3 min	537 MB (5,297s logged)

TABLE II: Computational profiling and hardware resource utilization on NVIDIA GeForce RTX 5050 Laptop GPU.

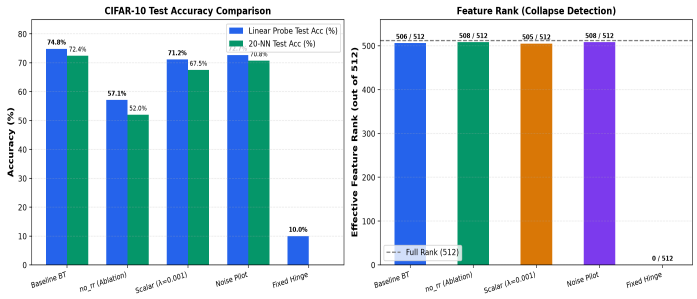


Fig. 2. Quantitative accuracy comparison and effective feature representation rank across Barlow Twins experimental variants.

VI. ABLATION STUDIES: ISOLATING REDUNDANCY REDUCTION

A. The Invariance-Only Ablation (no_rr): To determine whether cross-correlation decorrelation is strictly necessary, we ablated the redundancy reduction term (lambda = 0). The network was trained solely on diagonal invariance: L = sum_i (1 - C_ii)^2.

As reported in Table I and Fig. 2, eliminating redundancy reduction causes linear probe test accuracy to drop from 74.84% to 57.12%—a massive 17.72 percentage point drop. The 20-NN accuracy declined by 20.43 percentage points (to 51.98%).

Removing redundancy reduction substantially reduced downstream discriminability despite preserving high numerical feature rank (508/512). This shows that full rank alone does not guarantee semantically useful or sufficiently decorrelated representations. Batch standardization ensures non-zero variance per coordinate across the batch, but without off-diagonal decorrelation pressure, coordinates become redundant and functionally co-linear.

B. Static Hinge Thresholding and Rank-0 Collapse (fixed): In fixed_s0, a constant hinge threshold T = 0.05^2 was applied: L_rr = sum_{i!=j} ReLU(C_ij^2 - T), zeroing gradients when |C_ij| <= 0.05.

This negative control resulted in catastrophic failure: linear probe accuracy collapsed to 10.00% (pure random guessing), and effective feature rank fell to 0/512. The static-threshold configuration produced rank-0 collapse in this particular implementation and training setup. Without continuous repulsive gradients on small correlations, representations collapsed into a 1-dimensional singularity. In this reduced-scale setup, removing redundancy reduction substantially reduced downstream discriminability. The tested static-threshold

configuration produced rank-0 collapse, showing that this particular thresholding strategy failed to maintain a useful representation. Continuous redundancy reduction proved critical in this setup.

C. Regularization Attenuation (scalar): Weakening regularization by 5-fold (lambda = 0.001) degraded test accuracy by -3.69% (71.15%), showing that strong decorrelation pressure is essential for optimal feature separation.

D. Half-Batch Noise Estimation Pilot (noise): The noise-aware pilot underperformed standard Barlow Twins in this reduced-scale setup. Its adaptive threshold weakened some off-diagonal penalties; however, it remained stronger than globally reducing lambda (-2.15% to 72.69% vs -3.69% for scalar attenuation).

VII. DISCUSSION: MINIBATCH DYNAMICS & COLLAPSE

In canonical ImageNet training (N=2048), cross-correlation estimation standard error is sigma_err approx 1/sqrt(2048) approx 0.022. Under local consumer GPU settings (N=64), standard error expands to sigma_err approx 1/sqrt(64) = 0.125—nearly six times larger. Consequently, many small off-diagonal correlations observed in a given minibatch are transient sampling noise rather than persistent feature dependencies.

However, our negative control fixed_s0 proved that naively zeroing out small correlations causes immediate collapse. Thus, any successful small-batch adaptation must satisfy two fundamental conditions:
1. *Mass Conservation:* The total off-diagonal regularizing mass must equal 1 on average across all coordinate pairs.
2. *Continuous Gradient Flow:* Gradients must never be truncated or zeroed by discontinuous thresholds.

A. Small-Batch Gradient Variance & Normalization Sensitivity: At batch size N=64, stochastic fluctuations in empirical batch mean mu_i and variance sigma_i^2 increase significantly. When gradients backpropagate through standardization, small variances in denominator sqrt(sigma_i^2 + eps) can cause gradient scaling spikes. Enforcing gradient clipping at max_norm=5.0 and performing correlation math in FP32 ensures complete numerical stability without AMP scaler backoffs.

B. Comparison with Asymmetric and Contrastive Paradigms: Unlike SimCLR which requires large negative pools, or BYOL/SimSiam which rely on stop-gradients, Barlow Twins operates symmetrically. The off-diagonal term acts as an algebraic repulsive field: every coordinate pair repels if correlated, guaranteeing dimensional expansion without architectural asymmetries.

C. Representation Rank vs. Semantic Discriminability: A key theoretical insight from our empirical evaluation is the decoupling between mathematical rank and semantic utility. While fixed_s0 suffered complete collapse (rank 0), no_rr_s0 retained high rank (508) yet lost 17.72% accuracy. This confirms that decorrelation is not merely an anti-collapse safeguard, but an active driver of semantic factor disentanglement.

VIII. LIMITATIONS AND FUTURE WORK

While this study establishes an accessible reduced-scale baseline and isolates the critical necessity of redundancy reduction, several key avenues remain for future exploration:

1. Multi-Seed Replication: Although individual runs converged stably, expanding the empirical comparison to multi-seed ensembles with paired statistical hypothesis testing represents a valuable next step.

2. Extended Training Schedules: Canonical Barlow Twins benefits from extended pretraining (up to 1,000 epochs). Investigating 100- to 200-epoch schedules on CIFAR-10 could narrow the gap to full-scale benchmarks.

3. Cross-Dataset Generalization: Evaluating the learned representations on fine-grained benchmarks such as CIFAR-100 and TinyImageNet will provide broader insights into feature transferability.

4. Noise-Resilient Decorrelation: Developing continuous, mass-conserving reliability weighting schemes to adaptively attenuate small-batch correlation estimation noise without threshold-induced collapse remains a promising theoretical direction.

IX. CONCLUSION & KEY TAKEAWAYS

This paper presented a complete, rigorous reduced-scale reproduction and ablation study of Barlow Twins on CIFAR-10. Our findings establish three key scientific conclusions:

1. Consumer-Grade Accessibility: Barlow Twins reproduces robustly on consumer hardware, achieving 74.84% linear probe test accuracy and 72.41% k-NN accuracy in 120.2 minutes total pretraining time on an RTX 5050 Laptop GPU, dispelling the myth that self-supervised pretraining requires multi-node GPU clusters.

2. Redundancy Reduction is Indispensable: Completely eliminating the off-diagonal penalty (no_rr) induces a catastrophic 17.72 percentage point drop in linear probe accuracy, proving that view invariance alone is fundamentally insufficient for representation learning.

3. Continuous Repulsion Prevents Collapse: Static hinge thresholding precipitates catastrophic rank-0 collapse to 10.00% chance accuracy, demonstrating that continuous gradient repulsion across all coordinates is essential for dimensional expansion.

X. REPRODUCIBILITY & ARTIFACT AVAILABILITY

All code, configuration files, and evaluation scripts have been packaged into a self-contained, reproducible distribution. The repository includes automated unit tests verifying loss computations and autograd mechanics, as well as an interactive Google Colab notebook supporting zero-configuration reproduction on cloud T4 GPUs.

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